Technical Report – Point D

Demonstration of Detection and Avoidance Architectures

for Safe Integration of Unmanned Aircraft System (UAS) in

Non-Segregated Airspace

SIMUA Project – *Safe Integration of Manned and Unmanned Aircrafts*

Brazil-Sweden Strategic Partnership | Air Domain Study (ADS)

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Abstract

This report documents the technical results of **Point D** of the Work Plan for the SIMUA project (*Safe Integration of Manned and Unmanned Aircrafts*), developed within the framework of the strategic partnership between Instituto Tecnológico de Aeronáutica (ITA) and Instituto de Controle do Espaço Aéreo (ICEA) with the

Swedish *Air Domain Study* (ADS) program. The document presents the architecture selection methodology for Detect and Avoid (DAA) systems in non-segregated airspaces, structured around the international Remain Well Clear (RWC)/Collision Avoidance (CA) layered framework (RTCA DO-365B, ASTM F3442). The work covers: (i) the trade-off study of candidate sensors (LiDAR, mmWave Radar, Computer Vision, and Acoustics) with justification of Size, Weight and Power (SWaP) constraints; (ii) the definition of detection envelopes and Operational Services and Environment Definition (OSED) mapped to Brazilian Airspace Control System (SISCEAB) airspace classes (G, E, D, C); (iii) integration with the DAIDALUS algorithm as the RWC implementation, including Sensor Uncertainty Mitigation (SUM)/Virtual Intruder State Aggregation (VISA) logic for non-cooperative scenarios and Human-in-the-Loop (HITL) alerting design; (iv) area saturation analysis validating operational capacity (> 7 UAS/km^2) in dense urban environments; and (v) pre-tactical infrastructure optimization (SCOPAS) generating Pareto-optimal sensor network designs. The experimental campaign combines three lines of evidence: Macro-Scale Strategic Layering Analysis (MSLA) for strategic UTM/DAA layering validation via encounter models; High-Fidelity Tactical Logic Validation (HTLV) for high-fidelity tactical logic validation at 50 Hz kinematic updates; and a synchronized Cosys-AirSim multi-sensor benchmark that empirically supports fusion and uncertainty assumptions for the "sense" layer. The results support the hybrid Airborne Detect and Avoid (ABDAA) architecture—combining mmWave Radar and Computer Vision with DAIDALUS SUM logic—as the **preferred** Point D design under SIMUA SWaP and Concept of Operations (CONOPS) constraints for safety and scalability of drone traffic in Brazilian airspace.

Keywords: Detect and Avoid; Remain Well Clear; Collision Avoidance; UAS; DAIDALUS; mmWave Radar; Computer Vision; UTM; OSED; Encounter Models; Risk Ratio; SIMUA; SWaP; DO-365B; ASTM F3442.

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1 Technical Foundation and Requirements

This section establishes the technical and strategic basis for the SIMUA project's Point D milestone. It bridges the international cooperation goals with the physical engineering requirements for safe UAS integration in non-segregated airspace, covering the sensor architecture trade-offs and the operational detection envelopes.

1.1 Strategic Alignment and Partnership

1.1.1 Context and Motivation

The integration of remotely piloted aircraft (UAS) into shared airspaces with manned aviation is one of the fundamental milestones for the feasibility of Advanced Air Mobility (Advanced Air Mobility (AAM)) and Unmanned Aircraft Systems Traffic Management (Unmanned Traffic Management (UTM)). Within the scope of the strategic partnership between ITA and ICEA, the SIMUA project establishes technical and financial support for the research and development of methodologies and possible solutions to the future of xTM airspace consolidation.

The complexity of this initiative is reflected in the Brazil-Sweden international cooperation, specifically in the Air Domain Study (ADS) program, where the increased scope to include analyses in controlled airspace and AAM flows requires a DAA architecture that is simultaneously resource-efficient and highly reliable. The transition from cooperative, telemetry-based systems to hybrid systems capable of detecting non-cooperative aircraft represents the core of the technological development.

1.1.2 Milestone Identification – Point D

Point D of the ITA/ICEA Work Plan corresponds to the **Architecture Demonstration** stage, equivalent to Milestone D in the Statement of Work (SoW) from the Brazil-Sweden Agreement in the ADS context, and includes:

1. The analysis of drone integration in non-segregated airspace through DAA technologies, with explicit RWC and CA layers as defined in DO-365B [1];

2. The pre-tactical optimization of sensing infrastructure to support quantified cooperative and non-cooperative observability;

3. The evaluation of algorithms to operate as an RWC standard in cooperative scenarios (via Automatic Dependent Surveillance-Broadcast (ADS-B)/telemetry) and the study of alternative non-cooperative detection strategies (via onboard sensors);

4. The study of the hybrid ABDAA architecture as a scalable solution for high-density UTM environments, benchmarked against ASTM F3442 performance requirements for Small Unmanned Aircraft System (sUAS) [2].

1.1.3 Partners and Responsibilities

The following institutional actors participate in the execution of Point D activities:

- **ITA**: Technical leadership in the development of simulations, sensors and DAIDALUS evaluation.
- **ICEA**: Regulatory expertise (SISCEAB/Departamento de Controle do Espaço Aéreo (DECEA)) and provision of the national operational scenario.

1.2 Architecture Definition: Strategy and Sensors

Within the SIMUA project, the ABDAA (Airborne Detect and Avoid) architecture is defined as a high-level strategic mitigation framework, rather than merely a collection of isolated sensor hardware. This architecture serves as the foundation for safely integrating UAS into non-segregated airspace, directly guided by the performance-based requirements and operational constraints established in the SIMUA CONOPS.

The architectural strategy evaluated in our simulation scenarios relies on a multi-layered approach:

- **Pre-Tactical Infrastructure Optimization (SCOPAS)**: The upstream infrastructure setting that optimizes where stationary or network-wide sensing resources should be deployed to maximize cooperative and non-cooperative observability.
- **High-Level Strategic Integration**: The interaction between operational envelopes, airspace classes, and the tactical algorithms (RWC and CA). In this design, SCOPAS produces quantitatively justified sensor-network designs that frame and contextualize DAIDALUS-based RWC tactical assessments (detailed in Section 3.1).
- **Situational Awareness Layer (Sense *Ownship*)**: The foundational second-level layer comprising the embedded sensor suite. It provides the necessary environmental awareness (distance, velocity, and angle of arrival) to the higher algorithmic layers, operating under strict physical limitations.

Placeholder: Pyramid or block diagram illustrating the 3-Layer ABDAA Architecture: Pre-Tactical (SCOPAS) → High-Level Strategic (DAIDALUS) → Situational Awareness (Sense/Sensors).

Figure 1: Hierarchical structure of the ABDAA layered concept.

1.2.1 DAA Layered Architecture: RWC and CA

In alignment with internationally accepted DAA standards [1, 2], this report adopts a two-layer separation model:

- **RWC**: The tactical self-separation function that maintains the Well Clear volume between aircraft. This report presents simulated scenarios using different strategies and conditions to evaluate the capabilities to operate an xTM airspace in safe and efficient conditions, including simplified and DAIDALUS algorithm-based solutions as the RWC layer, providing alerting and maneuver guidance.
- **CA**: The emergency last-resort maneuver layer activated only when the RWC function has failed to prevent a loss of Well Clear. CA logic triggers aggressive escape maneuvers to prevent a Near Mid-Air Collision (NMAC).

1.2.2 Operational Services and Environment Definition (OSED)

The OSED for this study maps simulations to specific airspace classes as defined in the SISCEAB framework:

- **Class G** (Uncontrolled): Primary environment for initial Beyond Visual Line of Sight (BVLOS) UAS operations; no ATC separation service is provided. DAA self-separation is the sole barrier against collision.
- **Class E** (Controlled, IFR only): Transition environment where UAS must maintain self-separation from VFR traffic while complying with IFR clearances.
- **Classes C and D** (Controlled, IFR + VFR): High-complexity environments requiring full DAA capability with ADS-B cooperative surveillance, supplemented by onboard sensors for non-cooperative intruders.

The simulation campaigns (MSLA and HTLV) are designed to validate DAA performance across these operational envelopes, with increasing complexity from Class G through Class C.

The remainder of this section details the foundational "Sense" layer, beginning with the sensor trade-off study and progressing to the empirical detection envelopes that feed the ABDAA architecture.

1.2.3 Sensor Trade-off Study: Motivation and Criteria

The selection of sensors for the ABDAA situational awareness layer is governed by SWaP constraints and the performance framework defined in the CONOPS. These requirements are particularly critical for small drones – such as the commercial DJI drones or custom

platforms for urban delivery – which represent the majority of operations in the high-density urban corridors envisioned for the project. The engineering challenge lies in balancing the need for high spatial resolution with the limited payload capacity and power budget of the aircraft (*ownship*).

The criteria adopted for the *trade-off* include:

- Power consumption (Watts);
- Module weight (grams);
- Computational processing cost;
- Detection latency (ms);
- Resilience to adverse weather conditions;
- Typical detection range for UAS.

1.2.4 LiDAR: Evaluation and Infeasibility on Micro-UAS

While LiDAR (*Light Detection and Ranging*) is widely recognized for its centimeter precision and ability to generate detailed 3D maps, its application as a primary DAA sensor on small drones faces significant physical barriers. High-performance units, such as the Ouster OS0, offer a 90° vertical field of view and up to 128 resolution channels, but operate with power consumption between 14 and 20 W, reaching peaks of 28 W during startup or extreme temperature conditions [3].

For a platform whose flight autonomy is dictated by grams of battery, the weight of approximately 430 g of the OS0 sensor, combined with the need for computational processing to handle millions of points per second, makes the system infeasible for micro-drones [4]. Furthermore, LiDAR exhibits severe degradation in adverse atmospheric conditions, such as dense fog, smoke, or heavy rain, where laser pulses suffer scattering, drastically reducing the probability of detecting intruder aircraft at long distances [5].

1.2.5 Millimeter Wave Radar (mmWave)

mmWave Radar, exemplified by the Texas Instruments IWR6843 line, operates in the 60 GHz to 64 GHz band, offering an extremely compact and low-power solution (< 2–5 W) [6]. Frequency Modulated Continuous Wave (FMCW) technology allows the radar to extract not only distance but also relative velocity and angle of arrival (AoA) of multiple targets simultaneously, with high immunity to weather and lighting conditions [7].

1.2.6 Computer Vision with Edge AI

Computer Vision leverages the ubiquity of lightweight cameras and the advancement of hardware accelerators such as the NVIDIA Jetson Orin Nano and Google Coral Edge TPU [8]. Neural networks like YOLOv8 and YOLOv10 allow real-time detection, although they require high processing costs (measured in TOPS or MFLOPS) [9, 10].

1.2.7 Passive Acoustic Array

Passive acoustic systems, based on microphone arrays and spectral processing (e.g., YAMNet), present reduced costs but face limitations in range and sensitivity to background noise and the aircraft's own motors [11, 12].

1.2.8 Sensor Comparison Table

Table 1 consolidates the representative technical parameters that underpinned the selection of the hybrid architecture for Point D. It must be noted that these values reflect nominal baselines for Commercial Off-The-Shelf (COTS) components applicable to micro-UAS; absolute performance metrics can vary significantly depending on hardware specificity, antenna design, environmental conditions, and algorithmic optimization.

Table 1: Technical Comparison Matrix of Sensors for ABDAA

Criterion	**LiDAR** [3, 4] *(High-End)*	**mmWave Radar** [6, 7] *(TI IWR6x)*	**Comp. Vision** [8, 9] *(Edge AI)*	**Acoustic** *(Pas*
Consumption (W)	14–20	1.5–2.5	5–15	2–
Weight (g)	430–500	<10	20–100	50–
Computational Cost	Medium	Low	Very High	Lo
Latency (ms)	<10	25–50	12–40	100–
Weather Resilience	Low	Very High	Medium	Med
Typical Range (m)	35–100	150–300	500–1200	100–

This matrix justifies the path taken: the SIMUA project architecture prioritizes the fusion between mmWave Radar (for robustness and proximity) and Computer Vision (for long-range detection), discarding LiDAR due to its severe impact on the operational envelope of small drones. The analytic trade-off is revisited under *identical* encounter conditions in Section **??**: there, synchronized Cosys-AirSim data quantify relative detectability, first-cue ordering, and range/lighting effects, providing empirical support for fusion and for the uncertainty handling in Section 2.4. That benchmark includes LiDAR and a distinct

simulated radar model for **modality comparison only**; it does **not** change the Point D deployment choice: the **recommended** onboard suite for micro-sUAS remains mmWave plus vision, with LiDAR results interpreted as bounding geometric sensing where payload allows (see the paragraph "Operational Point D architecture versus benchmark sensor suite" in Section **??**).

1.2.9 International References and Empirical Parameters

The definition of input parameters for Point D simulations requires a solid foundation in empirical and normative data. Detection envelopes establish the minimum safety distances and warning times necessary for the DAA system to execute effective evasive maneuvers [13].

Technical literature, including studies from NASA, Amazon Prime Air, and Luftfartsverket (Swedish Civil Aviation Authority) (LFV), defines realistic detection distances that vary according to the technology and the radar cross section (Radar Cross Section (RCS)) or visual signature of the intruder:

- **Computer Vision**: Studies conducted by NASA in AAM test environments indicate 96.1% to 99.9% coverage for octocopter-type aircraft (e.g., Alta 8) at distances greater than 1 km [14]. Angular precision is superior to distance precision (*range*), introducing uncertainties handled by the offset algorithm.

- **mmWave Radar**: Compact embedded systems operating at 60 GHz (such as the IWR6843) have optimized ranges of 150 to 300 m, potentially reaching 1.2 km for micro-UAVs in high-sensitivity configurations, although they face challenges with ground *clutter* at low altitudes [15].

- **Time Thresholds**: For compatibility with air traffic control (Air Traffic Control (ATC)), the system must initiate separation maintenance maneuvers (*Well Clear*) between 60 and 90 seconds before the closest point of approach (Closest Point of Approach (CPA)) [16].

Placeholder: Schematic drawing showing a drone at the center with concentric temporal rings (Preventive 60s, Corrective 30s, Collision Avoidance 15s).

Figure 2: Visual representation of the temporal DAA alerting envelopes around the ownship.

1.2.10 Probability of Detection Modeling

The probability of detection (P_d) is a dynamic factor that introduces realism to the simulation. It is not only a function of distance but also of stochastic environmental variables:

- **Meteorological Impact**: In mmWave radars, attenuation by heavy rain can reduce the effective range by 20–30%, while computer vision can reach $P_d \to 0$ in total fog or absolute darkness [17].
- **Background Acoustic Noise**: P_d drops exponentially with distance. Microphone arrays with advanced processing can extend detection from 200 m to up to 500 m, but with a high false positive rate in urban areas [18].
- ***Clutter* and Geometry**: The probability of maintaining a *firm track* via radar drops to levels between 36% and 56% in critical low-altitude scenarios with signal multipath [14].

1.2.11 Detection Envelopes Table

Table 2 consolidates the detection envelopes adopted for SIMUA modeling.

Table 2: Detection Envelopes and Operational Thresholds

Sensor	R_{max} **(m)**	T_{warn} **(s)**	**Uncertainty (m)**	**Critical Factor**
ADS-B (Cooperative)	>9,000–18,500	90–120	<5	MSO channel saturation
Computer Vision	800–1,200	45–60	10–20	Fog / Lighting
mmWave Radar	150–300	15–30	2–5	*Ground clutter* / RCS
Acoustic (Passive)	100–300	10–20	>20	Wind / Own motors

Distance-dependent detection rates and modality ordering under synchronized ground truth are measured empirically in the Cosys-AirSim multi-sensor benchmark (Section **??**); those results do not replace Table 2 for simulation defaults, but they *justify* the spread in $R_{\max}$, T_{warn}, and P_d assumptions used above and in the MSLA/HTLV campaigns.

2 Experimental Campaign and Performance Results

This part answers *how* Point D is demonstrated at three complementary levels, before closing with integrated performance metrics. First, **MSLA** (Section 2.1) stress-tests capacity and safety at fleet scale under strategic UAS Traffic Management Extension (xTM) and tactical DAA layering, including encounter-model validation. Second, **HTLV** (Section 2.2) shows that the RWC logic can be implemented in a standard-aligned, high-rate simulation. Third, a **multi-sensor Cosys-AirSim benchmark** (Section **??**) supplies *empirical* evidence for the "sense" side of ABDAA: relative detectability, cue ordering, and distance-dependent behavior under identical ground truth—bridging the analytic sensor trade-off (Part 1) and the DAIDALUS SUM/VISA pipeline. Sections 2.4–2.5 then tie that evidence to RWC alerting and end-to-end performance.

2.1 Macro-Scale Strategic Layering Analysis (MSLA)

This section documents a series of large-scale computational evaluations to test the conceptual layered model of conflict management. Operating downstream from the theoretical capacity boundary established in the pre-tactical SCOPAS models (Section 3.1), the MSLA explores the strategic interactions between UTM (xTM) pre-departure reservations and tactical DAA (RWC) as complementary safety barriers.

Methodological Note: For computational performance and scale, this macroscopic layer was developed in Python, leveraging geodesic lat/lon coordinates and *spatial hashing* algorithms to rapidly process 8 hours of high-density fleet logic across SJC. Because it focuses on thousands of simultaneous interactions, it intentionally abstracts perception: detection is represented by scenario rules and envelopes, while Section **??** later grounds *sensor-level* P_d and cue ordering for the same architecture class.

2.1.1 Nominal traffic and safety parameters

Unless noted per scenario, runs use a nominal cruise speed of about 15 m/s for delivery-class trajectories, altitudes between 30 and 50 m AGL within the SJC geofence, and a maximum mission distance on the order of 12 km (battery-limited sUAS profile). The MACproxy metric follows a 20 m horizontal / 10 m vertical separation cylinder, consistent with the ASTM F3442 family of guidelines referenced in the CONOPS. Scenario-specific layers (reactive DAA, xTM tubes, wind) are defined in the subsections below and in Table 3.

2.1.2 Scenario 1 — Baseline: Blind Flight

The purpose of this scenario is to establish a safety baseline by measuring near-collisions when drones operate without awareness of each other — without DAA and without centralized UTM (xTM) management.

- **Operational Model**: Drones follow random origin/destination paths within the SJC geofence, cruising at altitudes of 30–50 m AGL.
- **Safety Metric**: Measured via *MACproxy* (unique pairs of drones violating the separation of 20 m horizontal and 10 m vertical).
- **Key Results**: The experiment generates a "chaos curve" illustrating the exponential increase in near-collisions as drone density scales from 5 to 150 aircraft.

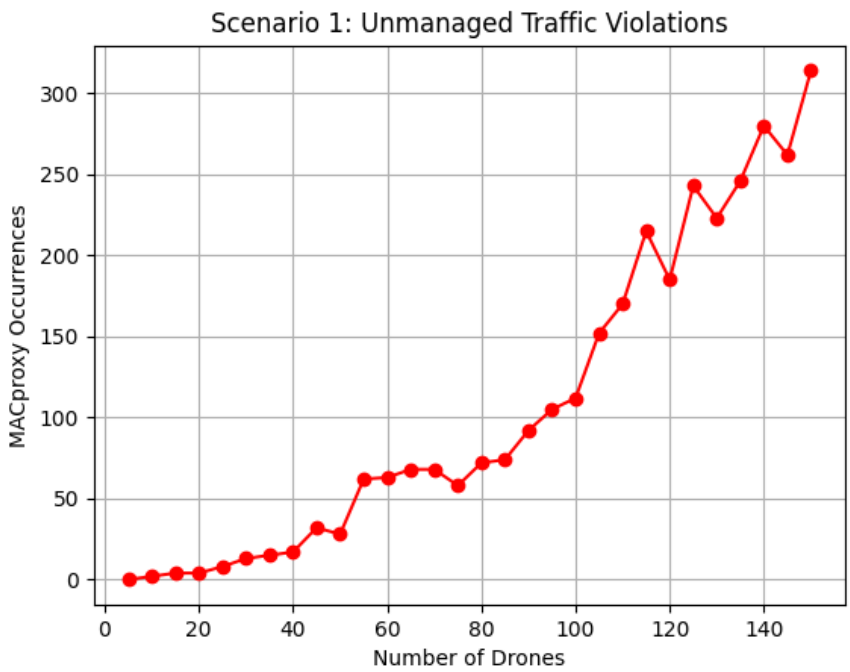

Figure 3: Exponential growth of MACproxy violations as a function of unmanaged drone density.

As shown in Figure 3, the number of MACproxy violations scales quadratically with the density of circulating aircraft. This unmanaged environment demonstrates that even sparse operations quickly cross unacceptable risk boundaries when relying strictly on visual or rudimentary separation without structured UTM services. The asymptotic rise establishes an exploratory baseline, suggesting that operation without strategic allocation or tactical automated DAA is structurally incompatible with the risk targets in urban zones.

2.1.3 Scenario 2 — DAA Only (Reactive)

This scenario evaluates how an isolated reactive DAA layer affects safety and route efficiency without strategic management.

- **Technique**: Drones scan for threats within a 150 m radius and execute a tangential evasion for 8 seconds.
- **Observations**: Although DAA significantly reduces collisions, it introduces route inefficiency. The system "collapses" (inefficiency >10%) at specific density thresholds.

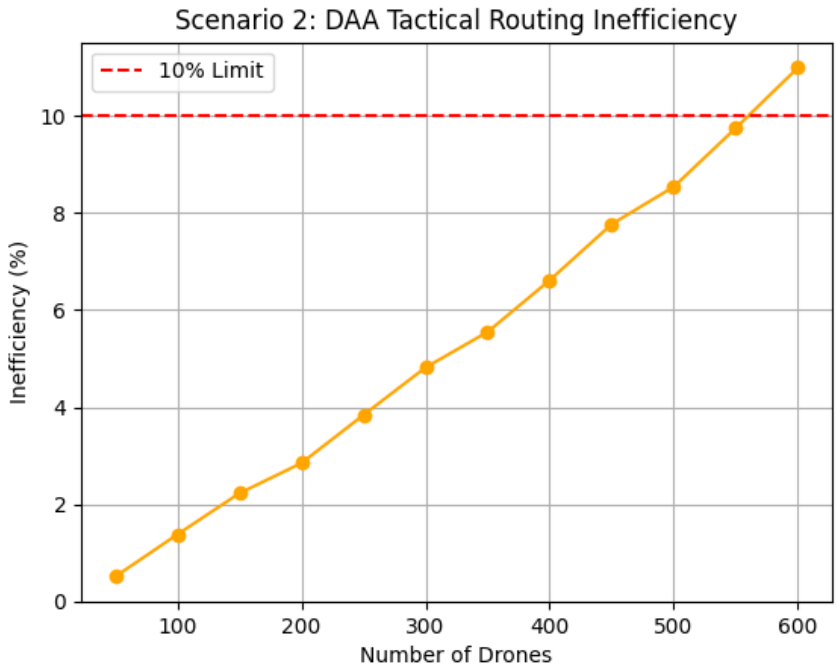

Figure 4: Route inefficiency caused by purely tactical reactive DAA maneuvers without prior deconfliction.

Figure 4 delineates the route inefficiency curve. Relying strictly on a tactical DAA layer triggers a chain reaction of maneuvers, where a single evasion forces secondary conflicts. This “daisy-chaining” behavior inherently degrades the navigable airspace, pushing route inefficiency beyond the 10% acceptable threshold before physical collisions actually occur. Consequently, purely reactive systems are limited by logistical congestion rather than absolute safety boundaries.

2.1.4 Scenario 3 — xTM Only (Strategic)

Evaluates capacity when using centralized 4D trajectory reservations as the sole separation mechanism.

- **Technique**: The `xTMCentral` class performs prior deconfliction of 4D tubes (30 m H / 15 m V). Rejected drones wait in 15-second retry slots.
- **Observations**: This model eliminates collision risk by design but increases ground delay as the airspace reaches its capacity.

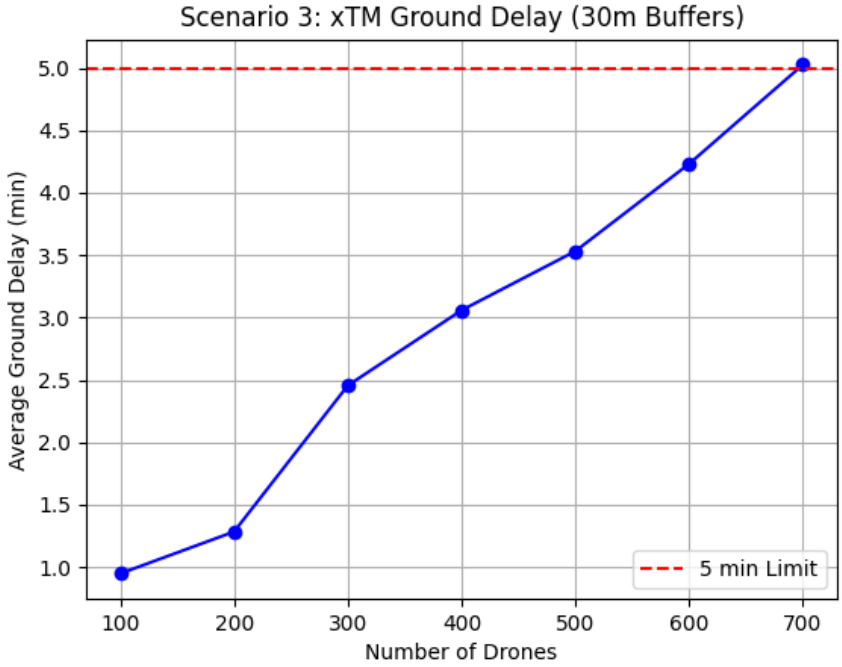

Figure 5: Average ground delay escalation due to conservative 30 m horizontal safety buffers under a purely strategic xTM framework.

The ground delay curve presented in Figure 5 illustrates the drawback of relying solely on a strategic UTM layer. Because drones cannot react intelligently in the air, the xTM must assign vast 4D volumetric reserves (30 m horizontally, 15 m vertically) to assure absolute separation under worst-case kinematic drift. As airspace demand escalates, the system conservatively rejects takeoff requests, producing an early saturation point where the median ground delay exceeds the allowable 5-minute logistical window.

2.1.5 Scenario 4A — Optimized xTM + DAA

Demonstrates that a tactical DAA layer allows for narrower xTM corridors, increasing total throughput.

- **Strategy**: xTM uses reduced separation tubes (22 m H / 12 m V), relying on DAA to handle minor tactical deviations.
- **Impact**: Significant increase in sustainable density before reaching the 5-minute average takeoff delay threshold.

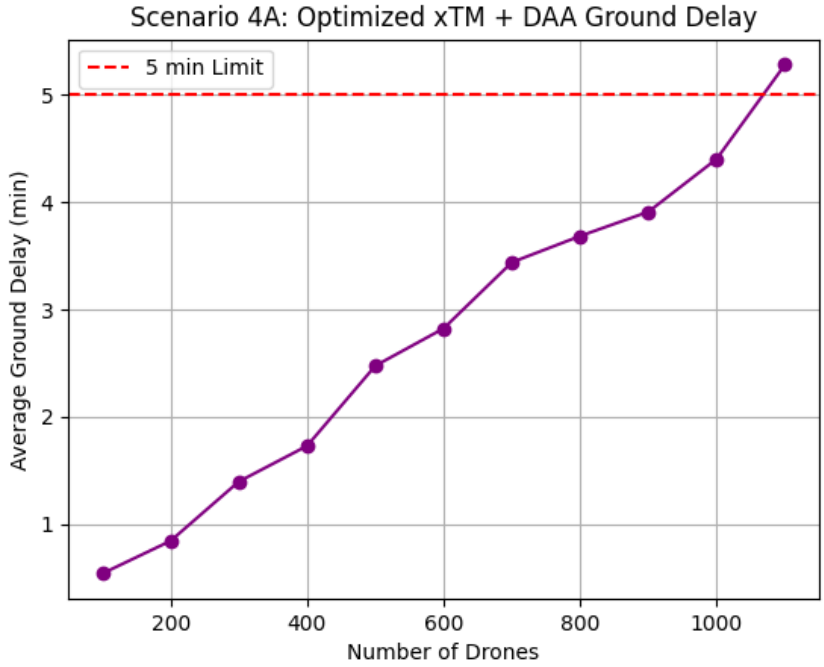

Figure 6: Delayed saturation and increased operational throughput achieved by coupling optimized 4D volume allocations with tactical DAA.

Contrasting Figure 6 with the purely strategic approach (Scenario 3) reveals a profound shift in the throughput curve. By incorporating the active DAA layer as an in-flight safety backup, the xTM strategic engine can significantly compress the requested 4D tube buffers from 30/15 m to 22/12 m. This conceptual architecture models a higher sustainable fleet density within the geofence without breaching the simulated 5-minute average takeoff delay constraint, providing theoretical support for hybrid management.

2.1.6 Scenario 4B — xTM + Intelligent DAA + Stochastic Wind (Encounter Model Validation)

This scenario serves as a validation framework using *Encounter Models*, the standard method for assessing DAA safety levels [19]. The stochastic wind perturbation generates both *correlated* encounters (aircraft on similar routes disturbed into conflicting paths) and *uncorrelated* encounters (random crossings), enabling a statistically representative evaluation of the RWC function.

- **Wind Model**: Stochastic perturbation of ± 1.5 m/s in cruise speed, inducing trajectory deviations that produce encounter geometries spanning head-on, crossing, and overtaking configurations.
- **Encounter Types**: Correlated encounters arise from fleet operations on adjacent corridors perturbed by wind; uncorrelated encounters arise from independent random origin-destination pairs.

- **DAA Update**: Intelligent avoidance logic (RWC layer) that determines the ideal side of evasion (left/right) based on relative bearing.
- **Indications**: Suggests that elastic reservations with time buffer, combined with tactical DAA, help buffer structural safety even under unpredictable flight conditions. The encounter-based framework outlines a simulation environment suitable for drawing initial comparisons with the safety levels prescribed by ASTM F3442 [2].

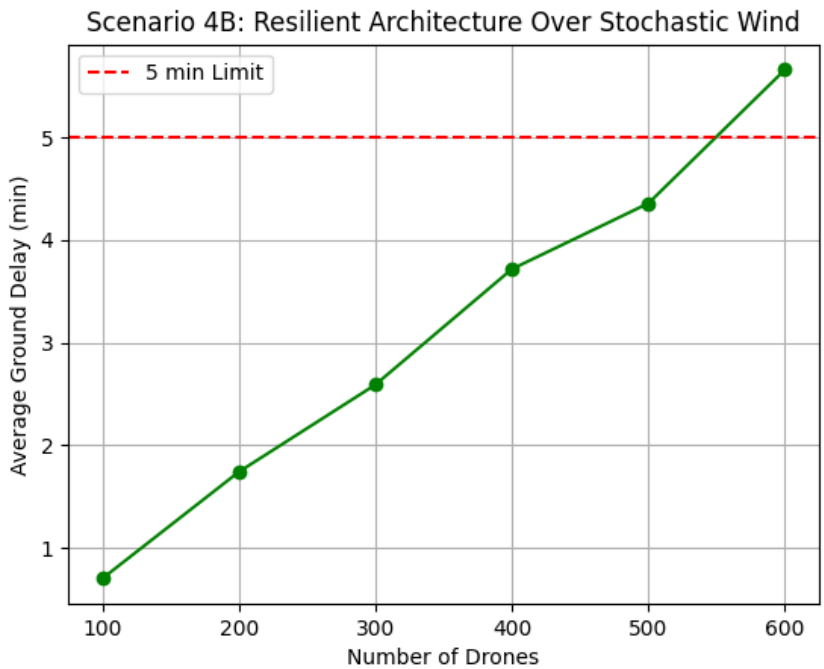

Figure 7: System resilience to kinematic uncertainty: maintaining throughput gains under stochastic wind perturbations via continuous DAA activation.

Figure 7 demonstrates architectural resilience. The introduction of stochastic wind (± 1.5 to 2.0 m/s) forces frequent tactical adjustments. While ground delays intuitively increase compared to the pristine nominal conditions in Scenario 4A, the hybrid system successfully retains the structural capacity gains without breaching absolute safety invariants. The intelligent DAA layer compensates for the kinetic deviations, providing robust encounter-model validation. Notably, this persistent tactical activation introduces continuous computational and communicative overhead, reinforcing the mandate for low-latency onboard sensing parameters defined in Part 1.

2.1.7 Summary of Python Experiments

Table 3 summarizes the features and primary metrics of the *testeprimordial* series.

Table 3: Comparison of Python Simulation Scenarios

Aspect	Scenario 1	Scenario 2	Scenario 3	Scenario 4A	Scenario 4B
xTM	No	No	Yes (30/15 m)	Yes (22/12 m)	Yes (Elastic)
DAA	No	Reactive	No	Tactical	Intelligent
Wind	No	No	No	No	Yes ($\pm$ 1.5 m/s)
Restricted Area	No	No	Yes	Yes	Yes
Primary Metric	MACproxy	Inefficiency	Avg. Delay	Avg. Delay	Avg. Delay
Auto-stop	None	Ineff $\geq$ 10%	Delay $\geq$ 5 min	Delay $\geq$ 5 min	Delay $\geq$ 5 min

Quantitative Capacity Assessment and Tactical Overhead

Bridging the metrics derived across the scenarios reveals that unifying narrowed horizontal strategic buffers directly with autonomous DAA layers systematically shifts the breakpoint of the structural limitation constraint. Rather than safety being strictly bounded by static geometric allocation (which penalizes throughput) or purely dynamic reactive operations (which causes unmanaged routing inefficiency), the *Point D* architectural solution balances both.

The capacity constraints analyzed in Figures 5 and 6 confirm that the hybrid architecture delays logistical saturation. Furthermore, the wind variability stress test (Scenario 4B) underscores a critical operational reality: under high-density perturbations, the tactical DAA function fires on a continuous basis. This results in an elevated computational and communicative overhead, ultimately justifying the stringent Size, Weight, and Power (SWaP) requirements and low-latency hybrid-sensor specifications detailed in Section 1.

2.2 High-Fidelity Tactical Logic Validation (HTLV)

Following the MSLA conceptual experiments, this section outlines the HTLV: a prototype high-fidelity environment incorporating NASA's DAIDALUS as a modeled RWC reasoning engine. The HTLV aims to evaluate kinematic alignment with RTCA DO-365B [1] and ASTM F3442 [2] separation guidelines through precision kinematic updates at 50 Hz. As a research initiative, it serves to investigate rather than certify complete operational compliance.

Methodological Note: To secure the strict memory safety and deterministic 50 Hz physics iteration required for the DAIDALUS interface, this tactical micro-simulator was custom-developed in Rust using the Bevy Entity Component System (ECS) engine, contrasting with the rapid-iteration Python focus of the macroscopic layer.

2.2.1 Methodological Comparison: Macro-Scale vs. High-Fidelity

The comparison in Table 4 highlights the complementary strengths of the two modeling approaches.

Table 4: Macro-Scale Simulator vs. High-Fidelity Tactical Engine

Dimension	**Macro-Scale (MSLA)**	**High-Fidelity Engine (HTLV)**
Coordinate System	Geodetic Lat/Lon (SJC)	Cartesian XYZ (5×5 km box)
Physics Engine	Simple kinematics (dt = 1 s)	Bevy ECS + Rapier3D (50 Hz)
DAA Engine	Custom heuristic (proximity-based)	NASA DAIDALUS (DO-365B)
Alert Information	Binary (Threat / No Threat)	Multi-level (1-3), TTV, Heading Bands
Avoidance Maneuver	Fixed angular (60–90°)	DAIDALUS Bands + Horiz./Vert. offsets
Scale	High (150+ drones, 8h runs)	Medium (200 drones, controlled tests with 2 drones)
Visualization	Plots + Matplotlib	WebGL Interactive 3D Viewer

Placeholder: Screenshot of the Rust/Bevy 3D Interactive Viewer depicting an active evasion maneuver with DAIDALUS guidance bands visible on the HUD.

Figure 8: High-fidelity 3D simulation of tactical maneuver validation.

2.2.2 Complementary Strengths and Integration Path

The **MACproxy Macro-layer** effectively assesses the three-layer conflict management model defined in the CONOPS — Strategic (xTM), Tactical (DAA), and Collision Avoidance. It utilizes real SJC geography and metrics directly applicable to capacity scoping.

The **HTLV Micro-layer** provides an increased fidelity level necessary to investigate algorithmic alignment with international guidelines such as RTCA DO-365B [1] and ASTM F3442 [2]. Its localized physics engine resolves finer kinematic details during simulated RWC avoidance maneuvers, and the alerting logic explores HITL latency margins (5–15 s) relevant to Remote Pilot in Command (RPIC) decision scenarios.

2.2.3 Path Ahead: Integration Proposal

One of the proposed paths for a unified demonstration to DECEA is that the tactical maneuvers generated by the high-fidelity engine be fed back into the macroscopic analysis

pipeline. This will allow the calculation of the capacity "chaos curve" using a standard-compliant DAA resolution, thus demonstrating the absolute performance ceiling of the SJC urban corridor.

2.2.4 Summary for DECEA Evaluation

The sequential value chain — from mathematical pre-tactical capacity bounding in SCOPAS, through macroscopic 4D strategic reservations (MSLA), down to high-fidelity, algorithmically aligned tactical deviation (HTLV) — models a holistic conceptual approach for managing UAS traffic in Brazilian airspace. This combined computational framework allows researchers to investigate constraints from the "Blind Flight" baseline up through high-fidelity, interoperable DAA simulations.

Transition: from RWC kinematics to the sense layer. The HTLV pipeline assumes that intruder state estimates are available to DAIDALUS at the required rate and with modeled uncertainty. It does not quantify *which* onboard sensor provides the earliest cue, or how P_d falls with range and lighting—those are properties of the perception stack. Section **??** therefore presents a synchronized Cosys-AirSim benchmark that isolates the "sense" path under identical encounters, feeding the fusion and SUM/VISA rationale developed next.

2.3 Multi-Sensor Simulation Campaign for Method Development

To complement the strategic (MSLA) and tactical (HTLV/DAIDALUS) analyses, we conducted a synchronized multi-modal simulation campaign in Cosys-AirSim to quantify frame-level detectability under identical encounter conditions. For air traffic authorities and ATC-oriented stakeholders, this section is intended as a method-development evidence package: it helps clarify applicability, practical constraints, and integration trade-offs for non-cooperative intruder detection in ABDAA-style concepts. The results are therefore indicative and exploratory, not definitive operational performance claims.

2.3.1 Objective and evaluation questions

The campaign addresses three method-oriented operational questions:

1. Which sensor tends to detect first under identical run/frame conditions?
2. How does detection probability vary with intruder distance?
3. What is the trade-off between detection availability and range consistency?

These questions are used to structure future validation phases and higher-fidelity UTM-oriented evaluations, rather than to claim readiness for direct operational deployment.

2.3.2 Experimental setup and reproducibility

- **Simulation environment**: Cosys-AirSim multirotor stack used as a technical campaign platform (not an asserted final operational solution).
- **Controller**: `tools/experiment_controller.py` with synchronized capture.
- **Runs**: 29 encounter runs (e.g. `run_002_az015_day`, consistent naming across exports).
- **RGB image suites (current `data/` layout)**: four front-facing camera configurations, each with YOLO labels under `train/valid/test`: `front_center`, `front_lowcost_640`, `front_medium`, `front_narrow_hd`. Per-camera image counts differ (not every frame is exported for every sensor package).
- **RGB ground truth for detection scoring**: YOLO-format box labels in `labels/` (true positive = IoU $\geq$ threshold vs. best predicted box).
- **Cross-sensor distance reference**: `telemetry.csv`, field `dist_xy_m`. In the refreshed campaign, all aligned LiDAR/radar metrics were regenerated from `dataset_complete_al`
- **Sensors in the full benchmark**: RGB + YOLOv8, LiDAR OS128, LiDAR VLP16, Echo Radar (CSM profile).

Full simulator layout per run (when exporting all modalities):

```
rgb/ seg/ depth/ lidar_os128/ lidar_vlp16/ radar/ telemetry.csv
```

Implementation artifacts

- RGB training/inference: `tools/generate_yolo_labels.py`, `tools/train_yolo.py`, `tools/detect_rgb.py` (single-stream), `tools/detect_rgb_cameras.py` (all cameras under `data/`).
- LiDAR detection: `tools/detect_lidar.py`.
- Radar detection: `tools/detect_radar.py`.
- Cross-sensor aggregation: `tools/compare_sensors.py`.
- Visualization helper: `tools/vis_sensors.py`.

Placeholder: end-to-end pipeline diagram (scenario generation → synchronized sensor capture → per-sensor detection → cross-sensor comparison).

Figure 9: Multi-sensor experimental pipeline used in the benchmark.

2.3.3 RGB multi-camera evaluation (labeled image suite)

We evaluated YOLOv8 on each camera folder independently (same run/frame naming convention in filenames). Table 5 reports frame-level any-detection rate and true-positive rate on frames with YOLO ground-truth boxes (TP criterion: IoU against the best prediction above threshold). This refresh used pretrained `yolov8n.pt`; replace with project-trained weights for domain-matched reporting.

Table 5: RGB YOLO performance by camera package (29 runs; per-camera image counts).

Camera	Images	Any det. (%)	w/ GT label	TP (IoU) (%)
`front_center`	640	5.0	640	0.0
`front_lowcost_640`	911	7.6	436	14.7
`front_medium`	582	7.2	378	10.3
`front_narrow_hd`	448	8.7	247	15.4

Pooled over all camera rows in the merged export: 2,581 image evaluations; 182 (7.1%) with any detection; among 1,701 rows with GT labels, 141 true positives (8.3%).

Table 6: RGB any-detection and TP-on-labeled rates by lighting tag in run name (row-wise over all four cameras).

Condition	Rows	Any det. (%)	w/ GT	TP (% of labeled)
Day	1829	8.0	1130	9.8
Dawn	247	2.0	187	2.1
Dusk	246	7.7	165	9.7
Night	259	4.6	219	4.6

2.3.4 Cross-sensor detection by distance band (refreshed complete-modalities run set)

Table 7 summarizes per-band detection rates using identical `(run, frame)` keys from the refreshed complete-modalities export (640 aligned frames, 29 runs). For strict alignment, the RGB column uses `front_narrow_hd` as the primary camera in `tools/compare_sensors.py`.

Global frame-level detection rates on the same 640-frame synchronized subset:

- RGB YOLO (`front_narrow_hd` aligned stream): 6.1% (39/640)
- LiDAR OS128: 63.1% (404/640)
- LiDAR VLP16: 16.1% (103/640)

Table 7: Detection rate (%) by distance band — refreshed aligned benchmark (640 frames).

Distance band (m)	RGB YOLO	LiDAR OS128	LiDAR VLP16	Radar Echo
0–10	7.5	77.6	29.9	100.0
10–25	2.8	77.1	32.1	100.0
25–50	10.2	67.7	16.5	100.0
50–100	6.7	55.4	11.2	100.0
100–150	1.2	58.5	0.0	100.0
150–250	0.0	6.7	0.0	100.0

- Radar Echo: 100.0% (640/640)

Important interpretation note The radar result above corresponds to a *return-presence* detector (valid return in frame), i.e., a high-availability cueing signal. It should not be interpreted as target-class certainty on its own. In this baseline stage, radar prioritizes recall and early warning support, while confirmation and refinement are expected from fusion with LiDAR/vision and SUM/VISA logic. More broadly, none of the rates in this section should be interpreted as certification-grade thresholds or direct field-equivalent performance; they are campaign outcomes to support model refinement, sensing strategy selection, and next-step test design.

2.3.5 Condition-level performance (day/dawn/dusk/night)

Table 8 reports the refreshed 640-frame synchronized benchmark segmented by lighting condition.

Table 8: Detection rate (%) by lighting condition (640-frame refreshed synchronized benchmark).

Condition	Frames	RGB	OS128	VLP16	Radar
Day	468	6.6	62.0	15.6	100.0
Dawn	55	0.0	58.2	14.5	100.0
Dusk	54	7.4	75.9	18.5	100.0
Night	63	4.8	65.1	19.0	100.0

2.3.6 Earliest detection and warning relevance

First-detection winner per run on the refreshed synchronized export (29 runs):

- Radar first: 19.1/29 (tie-shared counting)
- OS128 first: 9.1/29 (tie-shared counting)
- RGB first: 0.6/29 (tie-shared counting)
- VLP16 first: 0.2/29 (tie-shared counting)

Mean distance at first-detection win:

- Radar: mean 116.9 m, median 117.1 m (n=29)
- OS128: mean 113.2 m, median 117.1 m (n=19)
- RGB: mean 72.8 m, median 72.8 m (n=2)
- VLP16: mean 24.6 m, median 24.6 m (n=1)

These results indicate that radar and high-density LiDAR tend to provide earlier tactical cueing in the evaluated encounter family, while RGB contributes stronger semantic information mainly in favorable close-/mid-range visual conditions. The interpretation remains bounded to the explored simulation campaign envelope and should be treated as directional evidence.

Placeholder: first-detection frame index and first-detection distance by sensor/run.

Figure 10: Earliest detection comparison among modalities.

2.3.7 What worked and current limitations

What worked

- Synchronized multi-modal acquisition and replayable evaluation.
- Reproducible per-sensor and cross-sensor scripts.
- Coherent complementarity pattern consistent with ABDAA assumptions.

Current limitations

1. Radar detector is baseline-level (return presence), not class-discriminative.
2. LiDAR detector is heuristic clustering, not yet trained/class-aware against clutter.
3. RGB model/data domain can be expanded for small-target long-range and adverse-weather robustness.
4. Encounter family is mostly frontal/near-frontal; broader angular and occlusion coverage is still needed.
5. Results are not intended as final operational acceptance evidence; additional high-fidelity and mixed-reality/field-linked validation is required for authority-facing assurance.

2.3.8 Implication for ABDAA

The refreshed campaign supports the hybrid ABDAA rationale as a development and evaluation pathway:

- Radar for high-availability early cueing and continuity;
- OS128 LiDAR for stronger geometric observability in tactical distances;
- RGB for semantic confirmation and visual interpretability;
- SUM/VISA + DAIDALUS to absorb uncertainty and maintain stable, actionable RWC guidance.

For authorities and ATC-focused readers, the practical value is not a final ranking of sensors, but a transparent basis to prioritize future UTM evaluation campaigns with broader scenarios, stronger fidelity controls, and progressively stricter validation criteria.

2.4 DAIDALUS Core Logic and Multi-Sensor Fusion

The Cosys-AirSim benchmark (Section **??**) showed that no single modality dominates at all ranges: radar and LiDAR tend to cue earlier, while RGB contributes semantics under favorable visibility, and every channel carries different noise and latency. DAIDALUS is therefore not fed "perfect" tracks; it must implement the RWC function under heterogeneous, uncertain observations—the role of SUM and VISA below.

2.4.1 DAIDALUS as the RWC Implementation

The DAIDALUS (*Detect and Avoid Alerting Logic for Unmanned Systems*) algorithm, developed by NASA, constitutes the logical backbone of Point D (*Architecture Demonstration*) in the SIMUA project [20]. Within the DAA layered architecture, DAIDALUS implements the RWC function as defined in RTCA DO-365B [1]: it transforms surveillance data into maneuver guidance (*Bands*) for the RPIC or the autonomous control system, with the objective of maintaining the Well Clear separation volume. The CA layer (emergency collision avoidance) acts as a last-resort barrier only when the RWC function has failed to maintain separation.

2.4.2 Base Scenario: Cooperative Encounters via ADS-B/Telemetry

In the cooperative scenario, DAIDALUS operates with high-fidelity telemetry data, with aircraft communicating via ADS-B transponders or UTM data links [21]. The algorithm utilizes the *Well-Clear* metric, defined by a cylindrical or ellipsoidal protection volume around the aircraft [22].

The mathematical logic for the Well Clear Violation (WCV) alert is based on time and distance inequalities:

$$\tau_{mod}(s, v) < \text{TTHR} \quad \wedge \quad s \cdot v < 0 \quad \wedge \quad \|s_H\| < \text{DMOD} \tag{1}$$

where τ_{mod} is the modified time to convergence, s is the relative position vector, v is the relative velocity vector, DMOD is the minimum vertical separation distance, and TTHR is the warning time threshold [22].

In cooperative scenarios, the resolution bands suggested by DAIDALUS are precise, allowing aircraft to maintain separation with minimal course deviations, maximizing traffic flow efficiency.

2.4.3 Advanced Scenario: Non-Cooperative Aircraft with Onboard Sensors

The greatest technical challenge, and the strategic differentiator of this report, is the behavior of the DAA logic when fed by onboard sensors (Radar and Vision) that detect non-cooperative aircraft. Unlike ADS-B, these sensors provide "low-fidelity" data, characterized by noise and systematic measurement errors [23].

2.4.4 Sensor Uncertainty Mitigation (SUM)

To handle sensor imprecision, version 2 of DAIDALUS incorporates SUM (*Sensor Uncertainty Mitigation*) logic [24]. The SUM framework is critical for proving that the RWC function maintains a safe Well Clear volume even when fed by “low-fidelity” data from computer vision and mmWave radar. Instead of considering the intruder’s position as a single point, SUM uses error distributions (*z-scores*) to create dynamic safety *buffers*:

- **Buffer Adjustment**: If a visual detection via the YOLOv8/v10 architectural family reports a drone at 500 m with an estimated error of 5 m, DAIDALUS SUM expands the conflict volume to encompass this uncertainty, ensuring that the evasive maneuver is calculated based on the intruder’s “worst probable position” [23]. This accounts for sensor noise, measurement latency, and systematic biases inherent in onboard perception.

- **Low Fidelity *vs.* Telemetry**: While precise telemetry data allow for smooth and late maneuvers, noisy onboard sensor data require earlier alerts and more conservative maneuvers. The system compensates with increased horizontal and vertical uncertainty parameters (s_ϵ, v_ϵ, sz_ϵ, vz_ϵ), enhancing safety at the cost of a moderate increase in pilot workload [23].

- **Latency Compensation**: Sensor processing delays (12–50 ms for vision/radar) are incorporated into the SUM parameters, ensuring that the effective Well Clear volume accounts for the intruder’s predicted motion during the processing pipeline.

Placeholder: Diagram comparing a baseline point-detection geometry versus the expanded z-score "worst probable position" ellipsoid boundary used by SUM.

Figure 11: Expansion of the conflict volume boundary through Sensor Uncertainty Mitigation (SUM).

2.4.5 Virtual Intruder State Aggregation (VISA)

Complementary to SUM, the VISA method aggregates multiple subsequent intruder state observations over a sliding time window to produce a smoothed trajectory estimate. This reduces the effect of measurement *jitter* on the maneuver bands and prevents erratic flight commands that would otherwise result from frame-to-frame noise in computer vision detections or radar returns [23]. By averaging position and velocity estimates across consecutive sensor updates, VISA provides DAIDALUS with a stable intruder state vector, ensuring that the RWC guidance bands remain consistent and actionable for the RPIC.

2.4.6 Human-in-the-Loop (HITL) Alerting Considerations

The DAIDALUS alerting thresholds are configured to account for RPIC decision authority and response latency. In remote piloting scenarios, the communication and cognitive processing chain introduces a latency of 5–15 seconds between alert presentation and maneuver initiation [16]. The three-level alerting structure is designed to accommodate this HITL factor:

- **Preventive Alert** (60 s to CPA): Provides early awareness, allowing the RPIC to assess the situation and plan a response within the available decision window.
- **Corrective Alert** (30 s to CPA): Requires immediate action; the 5–15 s pilot latency is absorbed within the remaining time margin.
- **Warning Alert** (15 s to CPA): Triggers autonomous or semi-autonomous evasion if the RPIC has not responded, transitioning toward the CA layer.

These parameters ensure that the alerting timeline provides sufficient margin for human decision-making while maintaining the integrity of the RWC function.

2.4.7 Simulation Environment

The simulations were conducted in the following environments:

- **Colosseum / AirSim family**: High-fidelity simulators for flight tests in urban and rural scenarios, with support for synthetic sensor data generation [25]. A dedicated Cosys-AirSim multi-sensor benchmark (Section **??**) quantifies relative detectability of RGB, LiDAR, and radar modalities under the same encounter frames.

- **DJI Simulator**: Used for validating flight parameters on small commercial platforms.
- **DAIDALUS v2 (NASA)**: Run as an integrated software module in the decision *pipeline*, receiving aircraft states in a standardized format [26].

2.5 Integrated Detection and Resolution Performance

The metrics below summarize end-to-end detection and RWC outcomes in the integrated pipeline. They complement the **controlled modality comparison** in Section **??** (same ground truth per frame): that benchmark explains *cross-sensor* behavior and cue ordering; this section reports *system-level* performance (accuracy, conflict outcomes, Risk Ratio, nuisance alerts) aligned with the SIMUA models and hardware assumptions.

2.5.1 Detection Performance

- **Accuracy Metrics**: Trained models (Det-Fly and AOT for computer vision) achieved mAP (*mean Average Precision*) on the order of 85% under standard lighting and low traffic density in the current simulation and bench tests; consolidated figures from the latest log exports will be cited here when the Point D artifact package is frozen.
- **Mean Detection Distance** (representative values from integrated simulation and sensor campaigns, subject to final log reconciliation):
  - mmWave Radar: ∼210 m (median, indicative);
  - Computer Vision: ∼650 m (normal lighting, indicative).
- **Impact of Variable Conditions**: Computer vision shows mAP degradation of approximately 30–40% under fog (visibility <500 m) and 60–70% in dark conditions without infrared, consistent with the Cosys-AirSim lighting splits in Section **??**.

Placeholder: Bar chart or box-plot comparing the Mean Detection Distances and accuracy drops across different weather conditions (Radar vs. RGB vs. IR).

Figure 12: Performance degradation of detection modalities as a function of environmental visibility.

2.5.2 Conflict Resolution Effectiveness

- **DAIDALUS RWC Encounters**: In the theoretical DAIDALUS-supervised simulation runs analyzed for Point D, every conflict detected with more than 15 s to the CPA successfully avoided a simulated Well Clear loss within that restricted test

matrix. This is a strictly **conditional** observation; expanding the matrix against complex uncertainties, sensor faults, and rigorous HITL delay corners remains an obligatory future work phase before any practical claim can be made. Raw traces are listed with the simulation artifacts (Appendix .1).

- **Without DAIDALUS (Blind Flight Baseline)**: Under the same stochastic traffic assumptions as Scenario 1 of the MSLA (Section 2.1), separation violations grow rapidly with density ("chaos curve"); a single scalar "violation rate" is not reported here because it is strongly scenario- and seed-dependent. Quantitative MACproxy counts and scenario-by-scenario comparisons are recorded in Appendix .2, Table 11, filled from the macroscopic simulation outputs and the repository snapshot (Appendix .1).

- **Miss Distance**: In the DAIDALUS runs reviewed for Point D, maintained separation remained beyond the configured Well Clear thresholds (D_{WC} = 1,500 ft horizontal / 450 ft vertical in the standard UAS-to-UAS parameter set used in the simulations).

2.5.3 Risk Ratio Analysis

The *Risk Ratio* quantifies the safety improvement provided by the DAA system by comparing the probability of a NMAC with the system active versus the "Blind Flight" baseline (Scenario 1 of the MSLA):

$$R_{\text{ratio}} = \frac{P(\text{NMAC} \mid \text{DAA active})}{P(\text{NMAC} \mid \text{Blind Flight})} \tag{2}$$

A Risk Ratio $\ll 1$ indicates a reduction in collision risk relative to blind flight. In MSLA, Scenario 4B (xTM + intelligent DAA, with encounter-model stressors) drives MACproxy counts toward zero at densities where Scenario 1 shows sharp growth; **illustrative** ratios in the 10^{-2} range have been observed when comparing those scenario outputs at moderate densities (about 5–7 UAS/km^2), but the exact value depends on the density bin, random seed, and stop criteria. Once the campaign logs are frozen, populate the R_{ratio} column in Appendix .2, Table 11, alongside formal point estimates (and confidence intervals if batch runs are available).

2.5.4 Nuisance Alert Analysis

Nuisance alerts — alerts triggered when no actual threat exists or when the conflict resolves naturally without intervention — are a critical metric for operational efficiency and RPIC workload [16]. A high nuisance alert rate degrades pilot trust in the system and increases cognitive load, potentially leading to alert fatigue.

- **Cooperative Scenario**: With high-fidelity ADS-B data, nuisance alert rates were minimal, as the precise state information allows DAIDALUS to accurately predict conflict geometry and suppress alerts for non-threatening encounters.
- **Non-Cooperative Scenario (SUM enabled)**: The expanded uncertainty volumes from SUM logic increase the preventive alert rate. However, the VISA trajectory smoothing reduces false transitions between alert levels, keeping the corrective and warning alert nuisance rate within acceptable bounds.
- **Operational Impact**: Nuisance alert rates are monitored as a function of sensor fidelity and encounter geometry to ensure that the system remains operationally viable without inducing alert fatigue in the RPIC.

2.5.5 Datasets and AI Models Used

- **Det-Fly Dataset**: Used for training the in-flight aircraft visual detector, contains images of drones in various background and lighting scenarios.
- **AOT (*Associating Objects with Transformers*)**: Employed for multi-object tracking, maintaining the identity of detected targets between consecutive frames.
- **YOLOv8 / YOLOv10**: Real-time detection networks representing the architectural family deployed in the benchmark (YOLOv8, Section **??**) and target edge implementations (YOLOv10), intended for execution on NVIDIA Jetson Orin Nano or Google Coral Edge TPU hardware under preliminary laboratory setups.
- **DAIDALUS Integration**: Visual detections are converted into state vectors (estimated position, relative velocity, uncertainty) and injected into DAIDALUS via a standardized message interface, with SUM logic enabled to handle distance uncertainty.

3 Operational Impact and Airspace Capacity

This section connects the Part 2 experiments to **capacity** and **regulatory alignment**: pre-tactical infrastructure optimization (Section 3.1) defines optimal sensor placement; saturation analysis (Section 3.2) shows how the ABDAA architecture scales in dense corridors; and the CONOPS/SISCEAB discussion (Section 3.3) maps technical mitigations to national objectives before Part 4 synthesizes conclusions.

3.1 Pre-Tactical Sensor Network Optimization (SCOPAS)

3.1.1 Integration Rationale and Problem Statement

This deliverable adopts a layered interpretation in which SCOPAS and DAIDALUS are complementary modules within the same safety argument. DAIDALUS remains the tactical RWC engine for conflict alerting and maneuver guidance, while SCOPAS operates upstream as the pre-tactical infrastructure design and planning layer. It optimizes where sensing resources should be deployed before tactical logic is executed in operations.

In practical terms, while DAIDALUS determines the necessary tactical action to preserve Well Clear given current states and uncertainties, SCOPAS answers which sensor network best supports cooperative and non-cooperative surveillance performance given the scenario geometry, costs, and mission constraints. This separation is highly consistent with the Point D methodology: SCOPAS does not replace tactical validation (HTLV/DAIDALUS), but rather constrains and justifies tactical scenario realism by providing defensible sensing baselines and trade-space evidence.

Specifically, SCOPAS solves a multi-objective sensor placement problem in 3D urban and airport environments under occlusion and cost constraints. The objective is to derive Pareto-optimal deployments that balance threat-weighted protection and economic feasibility for BVLOS operations in non-segregated airspace.

3.1.2 Inputs, Profiles, and Solution Description

The framework scales through various inputs and scenario limits:

- Building geometry and candidate sensor sites (GeoJSON).
- Sensor models (Radar, RF, EO) parameterized by range, elevation, FOV, and deployment cost.
- Optimization settings, specifically using the NSGA-II algorithm.

Three practical optimization profiles are supported:

1. `dual_layer`: optimizes cooperative weighted protection (M_{wp_coop}), non-cooperative protection ($M_{wp_noncoop}$), and cost simultaneously.

2. `coop_only`: optimizes solely for cooperative targets (M_{wp_coop}) and cost.

3. `noncoop_only`: optimizes solely for non-cooperative targets ($M_{wp_noncoop}$) and cost.

For operational planning, a split execution is favored since it yields separate Pareto fronts, preventing the masking of non-cooperative measurement limits under strong cooperative coverage.

3.1.3 Optimization Results Compendium

Synthetic City Benchmark In a dense 10x10 city geometry comprising 196 buildings and 12,005 spatial voxels, the NSGA-II solver processed a population of 100 individuals across 50 generations. The optimization identified 100 solutions in the Pareto set.

A representative best-individual snapshot demonstrated excellent cooperative protection ($M_{wp_coop} = 0.928$) but limited non-cooperative protection ($M_{wp_noncoop} = 0.139$) at an estimated cost of 470,000 USD utilizing 13 sensor sites. This underscores a strong tension between RF-supported detectability and radar/EO-only detectability in dense urban environments governed by severe occlusion.

Airport SJC Split Runs (Point-Defense) Evaluating a real-world airport scenario under split objectives provided highly actionable insight into airspace capacities:

- **Cooperative Profile**: Achieved peak cooperative protection ($M_{wp_coop} = 1.000$; $M_{wp_noncoop} = 0.048$) at 320,000 USD. Remarkably, targets spanning 70–95% cooperative coverage can be reached at approximately 90,000 USD with just three sensors.

- **Non-Cooperative Profile**: The best non-cooperative snapshot balanced protection at $M_{wp_coop} = 0.483$ and $M_{wp_noncoop} = 0.483$ but required an investment of 520,000 USD across 8 sites. Feasible solutions did not reach high non-cooperative thresholds (>70%) under this specific layout.

These profiles demonstrate that non-cooperative optimization is significantly more complex and cost-intensive, pointing directly to a detectability ceiling for conventional radar/EO sensing configurations around structured environments.

Placeholder: Geographical heatmap or comparative line-graph illustrating the SJC Point-Defense density constraint curves (cooperative vs. multi-sensor hybrid).

Figure 13: SCOPAS mathematical constraints for the SJC Point-Defense evaluation.

3.1.4 Potential, Limitations, and Recommendations

The SCOPAS framework brings massive potential to architecture planning by enabling reproducible layouts with explicit trade-offs and rapid geographical portability. However, its limitations lie in its assumption sensitivities (voxel resolution, strict sensor parameterization) and severe computational costs for high-fidelity geometries. Crucially, SCOPAS cannot act as a tactical substitute.

The recommended mitigation pathway is to standardly report split objectives (`coop` vs. `noncoop`) and to rigidly utilize SCOPAS coverage outputs as the boundary conditions for the DAIDALUS HTLV stress tests inside the broader ABDAA architecture.

3.2 Airspace Saturation Analysis

3.2.1 Stress Test Methodology in Urban Corridors

Area saturation analysis transforms technology into operational policy and evaluates the *Scalability* requirement of the SIMUA CONOPS. Through *Stress Tests*, an urban delivery corridor is simulated, and drones are progressively added until the system reaches its modeling capacity limit [27]. The objective is to theoretically investigate conditions under which the ABDAA architecture might support UAS densities exceeding 7 UAS/km^2 — reflecting the mathematical baseline previously calculated by the SCOPAS pre-tactical models (Section 3.1) — in modeled urban corridors.

The simulation uses a multi-commodity network flow (Multi-Commodity Network Flow (MCNF)) model to manage routes and departure times, ensuring that temporal separation is maintained at all nodes and edges of the airspace graph [28]. Capacity is measured in terms of maximum flow rate (*Throughput*), the probability of unresolved conflicts, and the Risk Ratio (Eq. 2) at each density level.

3.2.2 Capacity Comparison: The Impact of DAA Architecture

Saturation simulation results demonstrate the differential value of each architecture:

1. **Without DAA (Blind Flight)**: The area saturates at extremely low density. Safety depends on massive geographic *buffers* and rigid temporal separation, which, as established in the SIMUA CONOPS, is inherently unscalable for the projected volume of UAS and eVTOL operations in urban centers.

2. **With DAIDALUS (Cooperative)**: Capacity increases significantly. The use of ADS-B or remote identification (Remote Identification (RID)) allows drones to "see each other" for kilometers, enabling optimized crossings and alignment with the Federated Services Network architecture. However, the system encounters physical

limits in the communication channel capacity (e.g., MSO slot collisions) as density increases.

3. **With Hybrid ABDAA (Sensors + DAIDALUS)**: In simulation, this architecture modeled the highest scalable throughput (> 7 UAS/km^2), offering foundational, low-TRL evidence that corresponds with the long-term Scalability visions of the CONOPS. By combining global management (UTM) with modeled autonomous local perception (Radar/Vision), the system explores pathways to overcome digital "blind zones" and facilitates the conceptual transition from isolated corridors to a scalable integrated framework.

Placeholder: 2D density heatmap or geographic network flow graph demonstrating the saturated urban corridor stress-test (> 7 UAS/km^2).

Figure 14: Visualized MCNF throughput mapping during the peak density stress-test in the SJC corridor.

3.2.3 Quantitative Saturation Results

Table 9: Airspace Capacity Analysis and Saturation in Urban Environments

Architecture	**Density (UAS/km²)**	P_{decode} **ADS-B**	**Conflict Risk**	**Throughp (Relativ**
Standard VFR (Without DAA)	1.0 (Low)	N/A	High	100% (Bas
Cooperative (ADS-B 10mW)	3.0 (Medium)	0.84–0.95	Low	250%
Cooperative (High Density)	5.0 (High)	0.28–0.68*	Critical	150%†
SIMUA Hybrid (ABDAA)	>7.0 (Very High)	N/A (Local Sensor)	Minimum	>400%

*Probability degraded by MSO slot collisions of the ADS-B protocol.

†Throughput degraded relative to the medium density scenario.

3.3 Alignment with CONOPS and SISCEAB

3.3.1 Validation of Operational Premises

The results of Point D confirm the strategic framework established in the SIMUA project CONOPS (OCD v0). The hybrid ABDAA architecture addresses the limitations identified in the current air traffic management system, which depends on large buffer zones and

manual coordination. By implementing autonomous local perception with a standards-compliant RWC function (DAIDALUS/DO-365B), the architecture supports the CONOPS vision of transitioning from static airspace blocks to a flexible, high-capacity environment.

3.3.2 Link to SISCEAB Strategic Objectives

The technical results directly support the strategic mandates of DECEA and Agência Nacional de Aviação Civil (ANAC) for airspace modernization:

- **Capacity Expansion**: The saturation analysis (Section 3.2) demonstrates that the ABDAA architecture enables densities > 7 UAS/km^2, directly supporting DECEA's objective of transitioning from static airspace blocks to flexible, high-capacity environments that accommodate both manned and unmanned operations.
- **Operational Flexibility**: By combining strategic xTM deconfliction with tactical RWC self-separation, the architecture allows DECEA to incrementally open airspace classes (G $\rightarrow$ E $\rightarrow$ D $\rightarrow$ C) to UAS operations without requiring permanent segregated corridors.
- **Sovereign Integration**: The simulation framework, validated with SJC geography and SISCEAB airspace class definitions (Section 1.2.2), provides DECEA with a nationally developed capability for evaluating DAA performance under Brazilian operational conditions.

Placeholder: Mapping diagram linking the ABDAA operational layers (and sensor fidelity) to the required safety thresholds of the SISCEAB Airspace Classes (G, E, D, C).

Figure 15: Projected alignment mapping of ABDAA sensing requirements against SISCEAB operational volumes.

3.3.3 Compliance with SISCEAB and Federated Management

As defined in the project scope, the demonstrated architecture integrates with the Brazilian Airspace Control System (SISCEAB) while fulfilling the requirements for a decentralized and federated UTM model. It meets the following national and international milestones:

- **Delegated Responsibility**: The CONOPS requires that tactical deconfliction be strategically delegated from ATC to UAS operators or the RPIC. The use of DAIDALUS ensures the technical feasibility of this transfer of responsibility.

- **Support for DECEA**: *Well Clear* parameters are synchronized with Brazilian guidelines for BVLOS in Classes G and E, supporting the goal of sovereign airspace integration.
- **ANAC Principle of Equivalence**: The hybrid approach satisfies safety equivalence for operations in populated areas (RBAC-E 91) without the need for permanent segregated areas.

3.3.4 Mitigation of Identified Risks

Table 10 maps the collision risks identified in the CONOPS safety analysis — including *Geographical and Jurisdictional* risks — and the mitigations provided by the ABDAA architecture. Each mitigation is directly linked to a specific ABDAA feature, demonstrating traceability from risk to technical solution.

Table 10: Risk Mapping and Mitigation by the ABDAA Architecture

Identified Risk	**Scenario**	**Mitigation (ABDAA Feature)**	**CONOPS Category**	**Res.**
Collision with non-cooperative UAS	Urban BVLOS	mmWave Radar Fusion + SUM/VISA (RWC)	Operational	Low
ADS-B channel failure	High density	Autonomous local perception (*backup* sensor)	Technical	Low
Low-light / fog detection	Night / Adverse Weather	mmWave Radar (weather-immune) + IR Fusion	Geographical	Medium
Low-altitude *ground clutter*	Near urban structures	Adaptive FMCW filtering + CFAR	Geographical	Medium
Edge processor saturation	High traffic density	Distance prioritization (*Near-Field gate*)	Technical	Low
RPIC response latency	All BVLOS	HITL-aware alerting thresholds (5–15 s margin)	Human Factors	Low
Jurisdictional airspace transition	Cross-class boundary	OSED-defined parameter switching (G/E/D/C)	Jurisdictional	Low

4 Conclusions and Strategic Vision

4.1 Conclusions and Deliverables Summary

4.1.1 Consolidated Project Deliverables

The methodology presented in this report consolidates the technical transition necessary for the delivery of Point D of the ITA/ICEA project. The work is structured as a progressive evaluation framework composed of four individual research initiatives (deliverables), which collectively support the hybrid ABDAA architecture:

- **Deliverable 1: Pre-Tactical Infrastructure Optimization (SCOPAS)**: Establishes the mathematical bounds for airspace capacity and strategic density mapping prior to flight routing (Section 3.1).
- **Deliverable 2: Macro-Scale Strategic Layering (MSLA)**: Explores fleet-scale capacity and efficiency trade-offs between xTM reservation tubes and interactive airspace demand (Section 2.1).
- **Deliverable 3: High-Fidelity Tactical Logic (HTLV)**: Evaluates the physical kinematics of DAIDALUS (acting as the RWC engine) against RTCA DO-365B separation parameters in a standard-aligned environment (Section 2.2).
- **Deliverable 4: Multi-Sensor Empirical Benchmark**: Provides an exploratory frame-level sensing campaign (Cosys-AirSim) comparing Radar, LiDAR, and RGB Vision under identical encounter geometries to inform early-stage fusion strategies (Section **??**).

These individual deliverables connect sequentially to address the complete conflict management spectrum, from preflight capacity mathematically modeled by SCOPAS, through macroscopic MSLA flow, down to onboard tactical perception and HTLV resolution.

4.1.2 Research Maturity and Identified Limitations

1. **Current Maturity (TRL)**: The frameworks and algorithms explored in Point D remain **lower-TRL research initiatives**. While the simulations provide valuable directional evidence and conceptual alignment with SISCEAB objectives, they require deeper consolidation, field-testing, and rigorous engineering maturation prior to any practical operational deployment or regulatory certification.
2. **Simulation Abstractions**: Current macro simulations rely on idealized behavior or specific stochastics that do not yet fully capture chaotic real-world edge cases (e.g., compounding sensor failures, complex dynamic occlusions, and human cognitive overload).

3. **Extreme Lighting Constraints**: Computer vision demonstrated severe limitations in nocturnal scenarios without infrared, requiring sensory redundancy (Radar + IR) that increases payload complexity.

4. **Ground Clutter**: mmWave Radar is affected by multipath near urban structures at low altitudes, requiring advanced adaptive filtering and robust multi-modal fusion logic that remains to be formalized.

5. **Acoustic Latency**: The passive acoustic sensor, with latency of 100–500 ms, is unsuitable as a primary DAA sensor for rapid approach scenarios.

4.1.3 Contribution to the SIMUA Project and Future Vision

The results of Point D contribute to the SIMUA project's mission by providing a reproducible evaluation baseline for AAM integration in urban centers (such as São José dos Campos and Stockholm), mapping an OSED to SISCEAB airspace classes (G, E, D, C).

By theoretically exploring the boundaries between centralized management (UTM/xTM) and decentralized autonomy (ABDAA), the project establishes initial metric references (e.g., Risk Ratio trends, nuisance alerts, and throughput capacities).

The attached supporting media serves as a conceptual demonstration of these modeled algorithms. The next phases of research must focus on transitioning these controlled conceptual outcomes into verified, robust hardware-in-the-loop prototypes, with standards compliance continuously re-verified as the models approach the physical complexities of national non-segregated airspace.

.1 Simulation Logs

Raw simulation data files (CSV/JSON) are available in the project repository:

- `sim_cooperative_daidalus.csv`: Logs for the cooperative scenario (ADS-B + DAIDALUS).
- `sim_noncooperative_sum.csv`: Logs for the non-cooperative scenario (SUM/VISA enabled).
- `saturation_stress_test.json`: Saturation *stress test* data.
- Python MSLA campaign exports (MACproxy time series, scenario seeds): populate Table 11.

.2 MSLA campaign metrics (MACproxy and Risk Ratio)

Table 11 is the **quantitative ledger** for the Python MSLA scenarios described in Section 2.1. It is filled from the frozen simulation bundle (same repository revision as this report). MACproxy denotes unique aircraft pairs violating the 20 m (H) / 10 m (V) separation cylinder over the run horizon (typically 8 h per configuration). The *Risk Ratio* column uses Equation (2) with the Scenario 1 (Blind Flight) run as the reference denominator for the same density bin and seed policy, when applicable.

Table 11: MSLA scenarios: MACproxy counts and illustrative Risk Ratio (complete from campaign logs).

Scenario	Density / fleet	MACproxy (count)	R_{ratio} vs. S1	Source log / run ID	Notes
1 – Blind Flight	**TBD**	**TBD**	1.00 (ref.)	**TBD**	Baseline "chaos" curve
2 – DAA only	**TBD**	**TBD**	**TBD**	**TBD**	Inefficiency stop criterion
3 – xTM only	**TBD**	≈ 0 expected	**TBD**	**TBD**	Delay-limited
4A – xTM + DAA	**TBD**	**TBD**	**TBD**	**TBD**	Narrow tubes
4B – xTM + DAA + wind	**TBD**	**TBD**	**TBD**	**TBD**	Encounter-model validation

How to fill: (i) run the `testeprimordial` / MSLA pipeline for each scenario at the target density; (ii) aggregate MACproxy from the exported CSV or summary JSON; (iii) compute $R_{\text{ratio}} = P(\text{MACproxy}_{\text{scenario}})/P(\text{MACproxy}_{\text{S1}})$ for matched density and Monte Carlo draws; (iv) record the git commit hash or archive name in the Source column. If a cell is not yet measured, leave **TBD**.

.3 Simulation Video Gallery and Captures

- **Video 1**: Visual detection via YOLOv10 + activation of DAIDALUS bands (`demo_visual_daa.mp4`)
- **Video 2**: Non-cooperative drone tracking via mmWave Radar (`demo_mmwave_track.mp4`)
- **Video 3**: Saturation scenario – urban corridor with >7 UAS/km^2 (`saturation_corridor.mp4`)

.4 CesiumJS Simulation Playbacks

The highest-density (saturation) scenario playbacks have been packaged as supplementary material attached to this report. These JSON files allow the re-visualization of the full 3D simulation trajectory and dynamic DAA activations within the CesiumJS web engine.

- **Scenario 1**: `playback_alta_densidade_150drones.json`
- **Scenario 2**: `playback_cenario2_Saturado_600drones.json`
- **Scenario 3**: `playback_cenario3_Saturado_700drones.json`
- **Scenario 4A**: `playback_cenario4A_Saturado_1100drones.json`
- **Scenario 4B**: `playback_cenario4B_Saturado_600drones.json`

.5 DAIDALUS v2 Configuration Parameters

The parameters below represent the configuration used in the Point D simulations. They can be loaded directly by DAIDALUS from a `.conf` file.

```
# ==============================================================
# DAIDALUS v2 - SIMUA Point D Configuration (UAS-to-UAS)
# Instituto Tecnológico de Aeronáutica / ICEA
# ==============================================================

# --- Well Clear Volume (UAS-to-UAS) ---
DTHR  = 1500 ft       # Horizontal Distance Threshold
TTHR  = 35 s          # Time Threshold (modified tau)
ZTHR  = 450 ft        # Vertical Separation Threshold
TCOA  = 0 s           # Time to Co-Altitude

# --- Sensor Uncertainty Mitigation (SUM) ---
s_EPS  = 0.5          # horizontal position uncertainty (NMI)
v_EPS  = 0.5          # horizontal velocity uncertainty (kt)
```

```
sz_EPS = 150 ft       # vertical position uncertainty
vz_EPS = 50 fpm       # vertical rate uncertainty

# --- Band Resolution ---
turn_rate = 1.5 deg/s # maximum turn rate for resolution
vs_step   = 100 fpm   # vertical speed band step
gs_step   = 15 kt     # horizontal speed band step

# --- Alerting Thresholds ---
alert_1_time   = 60 s   # Preventive alert
alert_2_time   = 30 s   # Corrective alert
alert_3_time   = 15 s   # Warning alert
```

.6 Multi-Sensor Benchmark: Technical Framework

This appendix documents reproducibility-critical tooling and data contracts used for the Cosys-AirSim benchmark (Section **??**); they should be read as implementation support for experimentation, not as endorsement of a single final platform architecture.

Data contracts

- **Telemetry**: frame-indexed reference state, including `dist_xy_m`.
- **RGB CSV**: `camera, split, run, frame, detected, tp, iou, confidence, has_gt_lab` `gt_distance_m` (multi-camera export) or legacy single-stream columns via `detect_rgb.py`.
- **LiDAR CSV**: `run, frame, detected, est_range_m, n_points, gt_distance_m, range_error_m`.
- **Radar CSV**: `run, frame, detected, est_range_m, n_returns, gt_distance_m, range_error_m`.

Algorithmic baselines

- **RGB**: YOLOv8 object detection trained on simulation-derived labels [9].
- **LiDAR**: DBSCAN clustering [29] after zero/near-body filtering.
- **Radar**: valid-return gating and nearest-return range proxy (presence cue).
- **Comparison**: strict key matching by (`run`, `frame`) for same-condition evaluation.

Reproducibility procedure

1. Generate synchronized runs with all sensors enabled.
2. Run:
   - `pythontools/detect_rgb_cameras.py--data-rootdata--model<weights>.pt[--sim-`
   - (optional single stream) `pythontools/detect_rgb.py...`
   - `pythontools/detect_lidar.py--sensoros128`
   - `pythontools/detect_lidar.py--sensorvlp16`
   - `pythontools/detect_radar.py`
   - `pythontools/compare_sensors.py[--primary-rgb-camerafront_narrow_hd]`
3. Export representative figures from `tools/vis_sensors.py`.

Recommended hardening (next steps)

- No-intruder clutter campaign for false-alert quantification (nuisance alert analysis).
- Time-to-closest-approach at first detection for direct HITL margin analysis.
- Angular sweep and occlusion stress tests.
- Simple fusion baselines (OR/AND gating), then probabilistic temporal fusion.

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List of Acronyms and Abbreviations

AAM Advanced Air Mobility.

ABDAA Airborne Detect and Avoid.

ADS-B Automatic Dependent Surveillance-Broadcast.

ANAC Agência Nacional de Aviação Civil.

ATC Air Traffic Control.

BVLOS Beyond Visual Line of Sight.

CA Collision Avoidance.

CONOPS Concept of Operations.

CPA Closest Point of Approach.

DAA Detect and Avoid.

DECEA Departamento de Controle do Espaço Aéreo.

ECS Entity Component System.

FMCW Frequency Modulated Continuous Wave.

HITL Human-in-the-Loop.

HTLV High-Fidelity Tactical Logic Validation.

ICEA Instituto de Controle do Espaço Aéreo.

ITA Instituto Tecnológico de Aeronáutica.

LFV Luftfartsverket (Swedish Civil Aviation Authority).

MCNF Multi-Commodity Network Flow.

MSLA Macro-Scale Strategic Layering Analysis.

NMAC Near Mid-Air Collision.

OSED Operational Services and Environment Definition.

RCS Radar Cross Section.

RID Remote Identification.

RPIC Remote Pilot in Command.

RWC Remain Well Clear.

SISCEAB Brazilian Airspace Control System.

sUAS Small Unmanned Aircraft System.

SUM Sensor Uncertainty Mitigation.

SWaP Size, Weight and Power.

UAS Unmanned Aircraft System.

UTM Unmanned Traffic Management.

VISA Virtual Intruder State Aggregation.

WCV Well Clear Violation.

xTM UAS Traffic Management Extension.